# UNDECIDABILITY OF ALGEBRAIC NUMBERS WITH LOGARITHMIC WEIL HEIGHTS

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In this article we prove that the field of algebraic numbers equipped with Weil logarithmic heights is undecidable. This was stated as a footnote in [1] and the second author was asked about the proof by Arno Fehm on a conference in Oberwolfach, January 2026. After realising that the margin was too narrow to contain the proof, this note was written. The authors thank Arno Fehm for the question. No AI was used in the process. The material was presented by the second author on the Northeast Model Theory Day in New York, on April 2026.

To make the statement precise, we use a variant of the language of globally valued fields from [1].

**Definition 1.** Let $K$ be a field and $\Gamma$ be an ordered abelian group (written additively). A $\Gamma$-*valued global height* on $K$ is a collection of functions $h : K^n \to \Gamma \cup \{-\infty\}$, for every natural $n \geq 1$, satisfying the following axioms.

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|---|---|---|
| Height of zero: | | $h(0,\dots,0) = -\infty$ |
| Height of one: | | $h(1,1) = 0$ |
| Product formula: | $\forall x \in K, x \neq 0,$ | $h(x) = 0$ |
| Invariance: | $\forall x \in K^n, \forall \sigma \in \mathrm{Sym}_n,$ | $h(\sigma x) = h(x)$ |
| Additivity: | $\forall x \in K^n, \forall y \in K^m,$ | $h(x \otimes y) = h(x) + h(y)$ |
| Monotonicity: | $\forall x \in K^n, \forall y \in K,$ | $h(x) \leq h(x,y)$ |
| Triangle inequality: | $\forall x, y \in K^n, x + y \neq 0,$ | $h(x+y) \leq h(x,y) + h(2,1)$ |

where $(x_i)_i \otimes (y_j)_j = (x_i y_j)_{i,j}$ is the Segre embedding. We write $\mathrm{ht}(x) = h(x : 1)$ for $x \in K$. A field $K$ together with a $\Gamma$-valued global height is called a *($\Gamma$-valued) globally valued field*.

Let $\mathcal{L}$ be a two sorted language with one sort $S$ equipped with the field language, and another sort $G \cup \{-\infty\}$ equipped a constant $-\infty$ and with the language of a divisible ordered abelian group on $G$ (with function symbols to be interpreted as division by $N$), and a family of function symbols $h_n : S^n \to G \cup \{-\infty\}$ (as above, we usually skip $n$ in the notation).

**Example 2.** Let $K$ be a number field and consider a tuple $x_0, \dots, x_n \in K$ with not all $x_i$'s equal to zero. Let $P_K$ be the set of places on $K$ (equivalence classes of absolute values on $K$). The *Weil logarithmic height* of $x$ is defined as

$$h_{\mathrm{Weil}}(x) = \frac{1}{[K:\mathbb{Q}]} \sum_{v \in P_K} [K_v : \mathbb{Q}_v] \log \max_{i=0,\dots,n} |x_i|_v,$$

where $|\cdot|_v$ is the absolute value corresponding to a place $v$ on $K$ normalised so that either $|p|_v = p^{-1}$ for some prime $p$, or $|\cdot|_v$ extends the Euclidean absolute value on the rational numbers. We also put $h_{\mathrm{Weil}}(0,\dots,0) = -\infty$.

For $x \in \overline{\mathbb{Q}}^n$, the value $h_{\mathrm{Weil}}(x)$ is independent of a number field $K$ for which $x \in K^n$. It is easy to check that Weil logarithmic height satisfies the axioms from Definition 1.

A number $a \in \overline{\mathbb{Q}}$ is called an algebraic integer if there is an integer monic polynomial $P \in \mathbb{Z}[X]$ such that $P(a) = 0$. Also, we call $a$ totally real if its Galois orbit is contained in $\mathbb{R} \cap \overline{\mathbb{Q}}$ (after fixing an embedding $\overline{\mathbb{Q}} \subseteq \mathbb{C}$). Julia Robinson proved [2] that the ring of totally real algebraic integers is undecidable.

**Theorem 3.** The $\mathcal{L}$-theory of algebraic numbers $\overline{\mathbb{Q}}$ with the logarithmic Weil heights is undecidable. More precisely, the ring of totally real algebraic integers is universally definable (over $\varnothing$) there.

*Proof.* This is by Lemma 5 and Lemma 6. □

The statement below is an instance of [1, Lemma 4.5] but we provide a self-contained proof.

**Lemma 4.** Consider the first order language $\mathcal{L}_0 = \{0, (q\cdot)_{q\in\mathbb{Q}}, +, -, \min\}$ and let $t(\alpha_1, \dots, \alpha_n)$ be an $\mathcal{L}_0$-term. There is a $\varnothing$-definable function $f : S^n \to G$ in the language $\mathcal{L}$, such that for all $a_1, \dots, a_n \in \overline{\mathbb{Q}}^\times$ we have

$$f(a_1, \dots, a_n) = \frac{1}{[K : \mathbb{Q}]} \sum_{v\in P_K} [K_v : \mathbb{Q}_v] \cdot t\big(-\log|a_1|_v, \dots, -\log|a_n|_v\big),$$

where $K \subset \overline{\mathbb{Q}}$ is any number field containing $a_1, \dots, a_n$, absolute values $|\cdot|_v$ are as in Example 2, and we evaluate $t\big(-\log|a_1|_v, \dots, -\log|a_n|_v\big)$ in $(\mathbb{R}, 0, (q\cdot)_{q\in\mathbb{Q}}, +, -\min)$.

*Proof.* Consider the following statement:

(♡) Let $t(\alpha_1, \dots, \alpha_n)$ be an $\mathcal{L}_0$-term. There are monomials $p_1, \dots, p_k$, $q_1, \dots, q_l$ in variables $x_1, \dots, x_n$, and a natural number $N \geq 1$, such that for any absolute value $|\cdot|$ on a field $K$ and any $a = (a_1, \dots, a_n) \in (K^\times)^n$ the number $t\big(-\log|a_1|, \dots, -\log|a_n|\big)$ equals

$$\frac{1}{N}\Big(\max\big(\log|p_1(a)|, \dots, \log|p_k(a)|\big) - \max\big(\log|q_1(a)|, \dots, \log|q_l(a)|\big)\Big).$$

Note that it is enough to prove (♡) as then the $\mathcal{L}$-definable function

$$f(x) = \frac{1}{N}\Big(h(p_1(x), \dots, p_k(x)) - h(q_1(x), \dots, q_l(x))\Big)$$

satisfies the conclusion of the lemma.

We prove (♡) by induction on the complexity of the term $t$ and on the number of its variables $n$. Base of the induction follows from the multiplicativity of absolute values, and from the fact that $-\min(-\alpha, -\beta) = \max(\alpha, \beta)$ in any linearly ordered group. For the induction step we consider two cases. First, when $t' = q \cdot t$ or $t' = t_1 + t_2$, (♡) for $t'$ follows from multiplicativity of absolute values. Second, assume that $t' = \max(t_1, t_2)$ and that $t_i\big(-\log|a_1|, \dots, -\log|a_n|\big)$ equals to

$$\frac{1}{N_i}\Big(\max\big(\log|p_{i,1}(a)|, \dots, \log|p_{i,k_i}(a)|\big) - \max\big(\log|q_{i,1}(a)|, \dots, \log|q_{i,l_i}(a)|\big)\Big)$$

for all tuples $a \in (K^\times)^n$ as in (♡) and $i = 1, 2$ respectively. By the first step we can replace $t_i$'s by the same multiple, and thus we can without loss of generality assume that $N_1 = N_2 = 1$. Then we can present $t\big(-\log|a_1|, \dots, -\log|a_n|\big)$ as

$$\begin{aligned}&\max\{\log|p_{i,r}(a) \cdot q_{3-i,s}(a)| : i = 1, 2; 1 \leq r \leq k_i; 1 \leq s \leq l_{3-i}\}\\ &\quad - \max\{\log|q_{1,u}(a) \cdot q_{2,v}(a)| : 1 \leq u \leq l_1, 1 \leq v \leq l_2\}.\end{aligned}$$

□

We use the following notation

$$\int t\big(v(a_1), \dots, v(a_n)\big)dv := f(a_1, \dots, a_n),$$

where $f$ corresponds to an $\mathcal{L}_0$-term $t(\alpha_1, \dots, \alpha_n)$ as in Lemma 4. Furthermore, we use shortcuts $t^+ = \max(t, 0) := -\min(-t, 0)$ and $t^- = \min(t, 0)$.

**Lemma 5.** The ring of algebraic integers $\overline{\mathbb{Z}}$ is universally definable (over $\varnothing$) in the field of algebraic numbers $\overline{\mathbb{Q}}$.

*Proof.* Consider the formula $\psi(y)$ defined as the conjunction of

$$\int \min\big(v(y)^+, -v(2)^-\big)dv = 0$$

and

$$\int \min\big(v(y) - v(2)^-, 0\big)dv = 0.$$

Note that $\overline{\mathbb{Q}} \models \psi(b)$ if and only if for every absolute value $|\cdot|$ on $\overline{\mathbb{Q}}$: $|b| \leq \max(|2|, 1)$ and if $|2| > 1$, then $|b| \geq 1$.

Define the formula $\varphi(x)$ as

$$(\forall y)\Big(\psi(y) \to \int \min(-v(x)^-, v(y)^+)dv = 0\Big).$$

Then $\overline{\mathbb{Q}} \models \varphi(a)$ if and only if for all $b \in \overline{\mathbb{Q}}$ for which $\psi(b)$ holds, there is no absolute value $|\cdot|$ on $\overline{\mathbb{Q}}$ such that $|a| > 1$ and $|b| < 1$.

We claim that this is equivalent to $a \in \overline{\mathbb{Z}}$. First, assume that $a \in \overline{\mathbb{Z}}$ and $b \in \overline{\mathbb{Q}}$ satisfies $\psi$. If there was an absolute value $|\cdot|$ on $\overline{\mathbb{Q}}$ such that $|a| > 1$ and $|b| < 1$, then $|2| \leq 1$, so $|\cdot|$ is non-Archimedean and we get a contradiction. On the other hand, assume that $\overline{\mathbb{Q}} \models \varphi(a)$. Let $|\cdot|$ be a non-Archimedean absolute value on $\overline{\mathbb{Q}}$. Let $p$ be a prime such that $|p| < 1$ and consider $b \in \overline{\mathbb{Q}}$ satisfying $b^n = p$ for some natural $n$. Note that if $n$ is big enough, we have $\overline{\mathbb{Q}} \models \psi(b)$. As $|b| < 1$ by construction, we have $|a| \leq 1$. As $|\cdot|$ was an arbitrary non-Archimedean absolute value on $\overline{\mathbb{Q}}$, we get $a \in \overline{\mathbb{Z}}$. □

**Lemma 6.** The field of totally real numbers $\mathbb{Q}^{\mathrm{tr}}$ is a quantifier-free definable (over $\sqrt{-1}$) subset of the algebraic numbers $\overline{\mathbb{Q}}$.

*Proof.* We use Lemma 4 with $t(\alpha, \beta) = \min\big(-\min(\alpha, -\alpha), -\min(\beta, 0)\big)$. Let $f : S^2 \to G$ be the corresponding $\varnothing$-definable function in $\mathcal{L}$ and consider the formula in variable $x$ in the sort $S$:

$$f\Big(\frac{x-i}{x+i}, 2\Big) = 0.$$

A number $a \in \overline{\mathbb{Q}} \setminus \{i, -i\}$ satisfies this formula if and only if

$$\sum_{v \in P_K} [K_v : \mathbb{Q}_v] \cdot \min\Big(\Big|\log \frac{|a-i|_v}{|a+i|_v}\Big|, \max(\log |2|_v, 0)\Big) = 0,$$

where $K = \mathbb{Q}(a)$. In other words, for any place $v$ on $K$, if $|2|_v > 1$, then $|a-i|_v = |a+i|_v$. Thus, the above formula defines algebraic numbers $a$ for which the distance to $i$ and $-i$ is the same with respect to every Archimedean absolute value on $\mathbb{Q}(a)$, i.e., it defines totally real algebraic numbers $\mathbb{Q}^{\mathrm{tr}}$. □

**Question 7.** We end the paper by asking the following:

(1) What are isomorphism classes of $\mathcal{L}$-structures on a fixed number field $K$? In particular, is the Weil logarithmic height the unique non-trivial $\mathcal{L}$-structure on $\mathbb{Q}$?
(2) Are $\mathcal{L}$-structures on $\overline{\mathbb{Q}}$ related to Diophantine approximation?
(3) What about the function-field case?

Of course some of these questions may be possible to solve by a machine now and some of them are vague. We encourage the reader to interact with these problems in any way they find suitable with the hope that it inspires interesting ideas, either with a use of computer or without it.

## References

[1] Itaï Ben Yaacov, Pablo Destic, Ehud Hrushovski, and Michał Szachniewicz. *Globally valued fields: foundations*. 2024. arXiv: `2409.04570 [math.LO]`.

[2] Julia Robinson. "On the decision problem for algebraic rings". In: *Studies in mathematical analysis and related topics*. Vol. IV. Stanford Studies in Mathematics and Statistics. Stanford Univ. Press, Stanford, CA, 1962, pp. 297–304.